

Challenge (Habanero)

- Q1.** A hiker's altitude h in meters after t hours is modeled by $h(t) = -2(t - 3)^2 + 1000$. What is the vertex of this parabola?
(A) (2, 1000)
(B) (3, 1000)
(C) (4, 1000)
(D) (1, 1000)
(E) (5, 1000)
- Q2.** The fuel consumption F in liters of a vehicle after traveling x kilometers is given by $F(x) = 0.02(x - 50)^2 + 20$. What is the minimum fuel consumption?
(A) 18 liters
(B) 20 liters
(C) 22 liters
(D) 25 liters
(E) 30 liters
- Q3.** A park ranger tracks the population P of a species after t years, modeled by $P(t) = -5(t - 2)(t + 1)$. What is the t -intercept of this quadratic function?
(A) -1, 2
(B) 1, -2
(C) -2, 1
(D) 2, -1
(E) -1, -2
- Q4.** The cost C in dollars of maintaining a hiking trail after x months is given by $C(x) = 0.1(x - 10)^2 + 50$. What is the vertex form of this quadratic function?
(A) $C(x) = 0.1(x - 10)^2 + 50$
(B) $C(x) = 0.1(x + 10)^2 - 50$
(C) $C(x) = -0.1(x - 10)^2 + 50$
(D) $C(x) = 0.1(x - 5)^2 + 25$
(E) $C(x) = -0.1(x + 5)^2 - 25$
- Q5.** The depth d in meters of a lake after x meters from the shore is modeled by $d(x) = -0.2(x - 20)^2 + 10$. What is the x -intercept of this quadratic function?
(A) 10, 30
(B) 20, 40
(C) 0, 40
(D) 14, 26
(E) 26, 14
- Q6.** A group of friends want to go on a hot air balloon ride. The height h in meters of the balloon after t minutes is given by $h(t) = -0.5(t - 5)^2 + 200$. What is the maximum height reached by the balloon?
(A) 150 meters
(B) 200 meters
(C) 250 meters
(D) 300 meters
(E) 400 meters
- Q7.** A wildlife conservation organization tracks the population P of an endangered species after x years, modeled by $P(x) = -2(x - 3)(x + 2)$. What is the vertex form of this quadratic function?
(A) $P(x) = -2(x - 3)(x + 2)$
(B) $P(x) = -2(x - 2)(x + 3)$
(C) $P(x) = -2(x + 3)^2 - 2$
(D) $P(x) = -2(x - 0.5)^2 - 12$
(E) $P(x) = -2(x + 0.5)^2 + 12$
- Q8.** The temperature T in degrees Celsius at a certain depth x in meters is given by $T(x) = -0.01(x - 100)^2 + 20$. What is the temperature at $x = 50$ meters?
(A) 10
(B) 15
(C) 18
(D) 20
(E) 25

Open Ended Questions (FRQ)

- Q9.** The population P of a species of birds after t years is modeled by $P(t) = -3(t - 2)(t + 1)$. Find the t -intercepts, vertex, and axis of symmetry of this quadratic function. Show your work and explain your answer.
- Q10.** A company offers a x -kilometer hiking trip. The cost C in dollars of the trip is given by $C(x) = 0.05(x - 10)^2 + 200$. Find the vertex form of this quadratic function, and interpret the meaning of the vertex in the context of the problem. Show your work and explain your answer.

Chef's Tips

- Make sure students show their work for FRQs
- Emphasize the importance of interpreting parameters in vertex and intercept form
- Allow students to use graphing calculators to check their answers